PHL 1010: Electromagnetism and Optics

Course Instructor: Sunil K Khijwania

Office: 332, Department of Physics, IITJ

Contact: 1606
skhijwania@iitj.ac.in

Lecture Plan

Lecture 1	**Monday**	हिंदी	08:00 AM - 08:50 AM
		English	05:00 PM - 05:50 PM
Lecture 2	**Wednesday**	हिंदी	08:00 AM - 08:50 AM
		English	01:00 PM - 01:50 PM
Lecture 3	**Friday**	हिंदी	08:00 AM - 08:50 AM
		English	01:00 PM - 01:50 PM

Office Hours

Day	Time
Friday	5 – 6 PM

Asking questions in the class is encouraged.

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Hence, I strongly encourage you to come to me during my office hours, and to ask lots of questions (which you could not ask in the class) about the course material and its applications.

This gives me a chance to work with you individually, and to help you against any difficulty you may be facing related to the conceptual understanding, which you would have missed during the class hours.

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Tentative Exam Schedule/Scheme

Exam	Date	Contribution
Quiz 1	5 September	10%
Mid-sem	17 -20 September	30%
Quiz 2	31 October	10%
End-sem	20 -26 November	50%

No make-up for Quizzes and Mid-sem.

End-sem make-up according to IITJ regulations.

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Attendance

- **Attendance will be taken during every lecture class**
- **Any false signature would mean -5 to entire class**

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Text Book/Reference Material

Text Books

- **Introduction to Electrodynamics**
  David J Griffiths, Third Edition, Rs 195 only!
- **Optics**
  Ajoy Ghatak, Sixth Edition McGraw-Hill

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Reference Books

- **Engineering Electromagnetics**
  Ida, N., (2015), 3rd Edition, Springer Nature
- **Optics**
  Hecht, E., (2017), McGraw Hill Education India
- **Principle of Optics**
  Born, M. and Wolf, E., (1999), Cambridge University Press.

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Text Book/Reference Material

- **Keep an eye on Google Classroom**
  Google class-code: guvf32k3

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Course Content

- **Vector Calculus**
- **Electrodynamics** — **Electrostatics**, **Magnetostatics**, **Electrodynamics**
- **Optics**

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Introduction & Overview

Introduction & Overview

Why to study Electrodynamics ?

Why to Study Electrodynamics ?

To understand the Nature!

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Which finally culminates in the consideration/ understanding of all the existing forces in the universe:

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Which finally culminates in the consideration/ understanding of all the existing forces in the universe:

- can we stand/walk/write?

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Which finally culminates in the consideration/ understanding of all the existing forces in the universe:

- can we stand/walk/write?
- can planet occupy there position/orbit in the universe where they are?

18

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Simply NO

20

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- can we stand/walk/write?
- can planet occupy its position/orbit in the universe where they are?
- or can universe exist?

Simply NO

Forces are pivotal for the existence of the universe

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Why to Study Electrodynamics ?

Fundamental forces in the nature:

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Why to Study Electrodynamics ?

Fundamental forces in the nature:

- Gravitational force

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Fundamental forces in the nature:

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- Weak (nuclear) force

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but….

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Fundamental forces in the nature:

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but…. **….where is friction?**

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….where is chemical force?

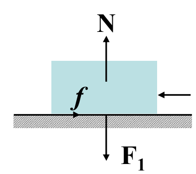
Why to Study Electrodynamics ?

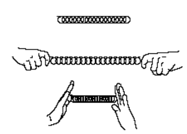
Fundamental forces in the nature:

- Gravitational force
- Weak (nuclear) force
- Electromagnetic force
- Strong (nuclear) force

but…. ….where is friction?
….where is normal force?
….where is chemical force?
….list goes endless…………

Why to Study Electrodynamics ?





Most of the forces we observe between macroscopic objects are complicated manifestations of basic EM interaction

Why to Study Electrodynamics ?

not because of gravity!

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all are connected to Electromagnetism or simply Electrodynamics

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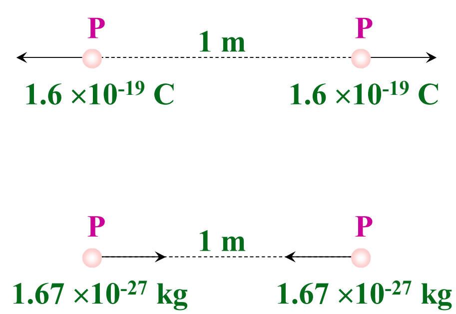
all are connected to Electromagnetism or simply Electrodynamics

its not an exaggeration to say that we live in an electromagnetic world!

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force



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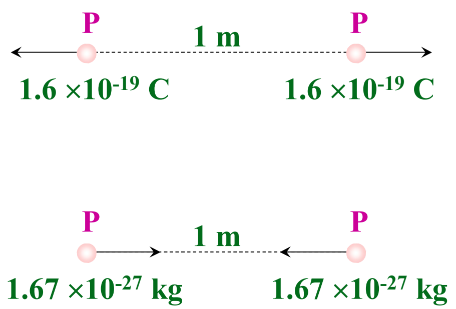

$$\frac{F_e}{F_g} \approx 10^{36}$$

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Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

well that's the comparative picture of electromagnetic force verses gravity….

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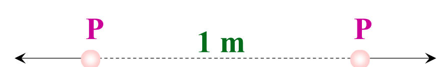

$$F_e = 2 \times 10^{-28}\,\mathrm{N}$$

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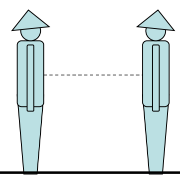

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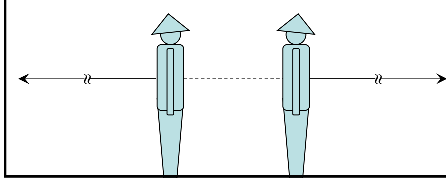


F would be enough to lift a weight equal to that of the earth!

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

Question

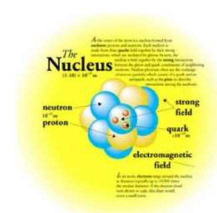
why don't we experience such strong force in our daily life?

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

Question

what holds nucleus together?



^{4}He: Two protons and two neutrons in a volume of radius 10^{-15}m

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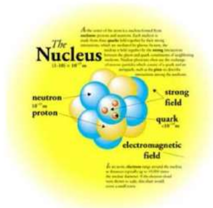

3. Strong (Nuclear) Force

Why to Study Electrodynamics ?

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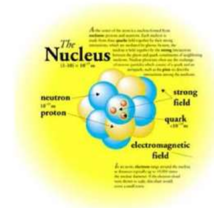


3. Strong (Nuclear) Force

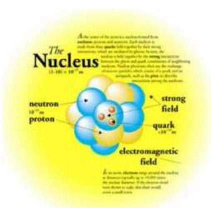
100 times more powerful than EM repulsion

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

Question



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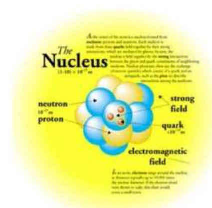
1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

Question

…is it charge dependent ?



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3. Strong (Nuclear) Force

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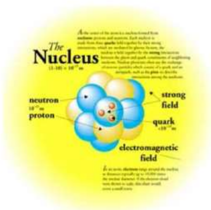

…is it charge dependent ?

…why do not we experience them ?

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

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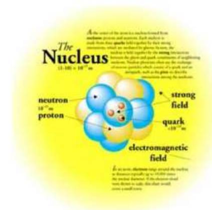


…is it charge dependent ?

…why do not we experience them ?

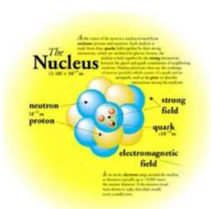
…why does not nucleus collapse ?

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

Answer



…is it charge dependent ?

charge in-dependent !

…why do not we experience them ?

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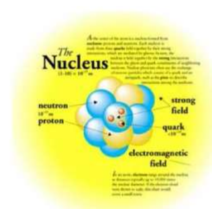
Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

Answer



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charge in-dependent !

…why do not we experience them ?

extremely short ranged !

…why does not nucleus collapse ?

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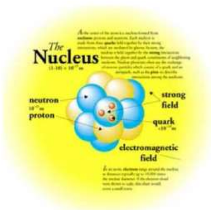
Answer



…is it charge dependent ?

charge in-dependent !

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extremely short ranged !

…why does not nucleus collapse ?

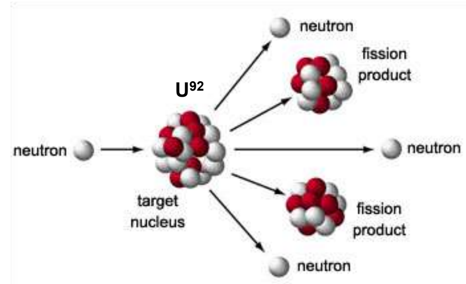
repulsive at very short ranged ~ 0.4×10^{-15} m !

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

Important Consequence

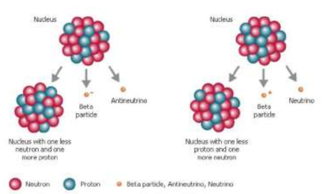


Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

consider β-decay



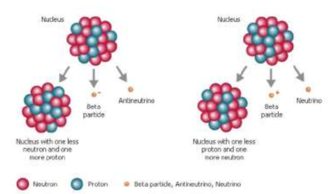
Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

4. Weak (Nuclear) Force
   acting between leptons and hadrons

consider β-decay



Why to Study Electrodynamics ?

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3. Strong (Nuclear) Force

4. Weak (Nuclear) Force

Why to Study Electrodynamics ?

1. Gravity & Electromagnetic Force

3. Strong (Nuclear) Force

4. Weak (Nuclear) Force

How do they affect our lives and help us in understanding the universe/nature?

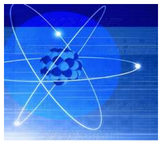
Why to Study Electrodynamics ?

Optics

Why to Study Electrodynamics ?

Electrical Engineering

Why to Study Electrodynamics ?



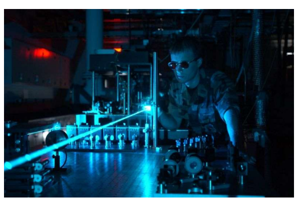

Atomic/Nuclear Engineering

Why to Study Electrodynamics ?



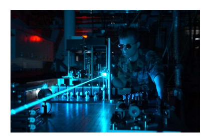

Communication Engineering

Why to Study Electrodynamics ?




Home Land security

Why to Study Electrodynamics ?




Optical Communication….

Why to Study Electrodynamics ?

are consequence of the manipulation of EM forces by nature itself….

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Why to Study Electrodynamics ?

are consequence of

human made

controlled

& precise

manipulation of

EM forces ….

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Brief History of Electrodynamics

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Brief History of Electrodynamics

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Brief History of Electrodynamics

600 B.C.

Amber

Brief History of Electrodynamics

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Amber

Loadstone

Brief History of Electrodynamics

600 B.C.

Amber

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Thales of Miletus

Brief History of Electrodynamics

Early Period (1600 A.D.)

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1600 A.D. Gilbert

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- +/- charge

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1785 A.D. Coulomb
- Measured Electrostatic force

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- Light has finite velocity

80

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Electricity & Magnetism developed separately

Brief History of Electrodynamics

Classical Period (1800-1900 A.D.)

1801 A.D. Young Double slit exp – light as a wave

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…at last determined relation between E & M

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Great unification of electricity and magnetism????

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Not Yet

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Phenomena being asymmetrical

Brief History of Electrodynamics

Classical Period (1800-1900 A.D.)

1831 A.D. Faraday

in 1812, 20 year old Faraday attended lecture of Humphry Davy, Director of Royel Institution

Brief History of Electrodynamics

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1831 A.D. Faraday
- **Law of EM induction, electrolysis**

Brief History of Electrodynamics

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….all being path breaking one!

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↓

could not unify the two phenomena into a single theory

Brief History of Electrodynamics

Classical Period (1800-1900 A.D.)

1864 A.D. Maxwell

- Vacuum polarization current (electric counterpart of Faraday's magnetic induction)
- EM Theory (field equations)

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Electricity + Magnetism ⟶ Electromagnetism *or simply* Electrodynamics

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Justification to Faraday's speculation about the EM nature of light

Brief History of Electrodynamics

Classical Period (1800-1900 A.D.)

1864 A.D. **Maxwell**
- **Vacuum polarization current** **(electric counterpart of Faraday's magnetic induction)**
- **EM Theory (field equations)**

1886 A.D. **Hertz**
Generation & detection of EM wave **(confirming Maxwell Theory)**

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1900 A.D.

Electricity + Magnetism ⟶ Electrodynamics
Optics ↗

Brief History of Electrodynamics

Modern Period (1900 – Present)

EM unification was a great event in science…

…for its beautiful simplification

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unification of matter, energy, momentum!

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unification of matter, energy, momentum!

unification of space & time!

Brief History of Electrodynamics

Modern Period (1900 – Present)

1905 Einstein

Special Theory of Relativity

Brief History of Electrodynamics

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Special Theory of Relativity

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- **Applied Relativity to cosmology**

Brief History of Electrodynamics

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- **Dreamed for further unification of gravity and ED…**

  …that way, relating all the four forces

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ED + QM + STR ⟶ Quantum ED

Brief History of Electrodynamics

Modern Period (1900 – Present)

Quantum ED

- Highly accurate description of nature in all its aspects from far inside the nucleus, to microelectronics, to table to chair, to the most remote galaxy

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Brief History of Electrodynamics

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Brief History of Electrodynamics

Modern Period (1900 – Present)

Quantum ED

- Highly accurate description of nature in all its aspects from far inside the nucleus, to microelectronics, to table to chair, to the most remote galaxy
- Precision of the agreement of QED with experiment in atomic/particle physics where RQM rules is astounding!
- Prediction of magnetic moment of electron has been proved correct to at least 12 significant digits!

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Brief History of Electrodynamics

Modern Period (1900 – Present)

1960 Glashow, Salam, Weinberg

ED + Weak Force $\longrightarrow$ Electroweak theory

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1960s

Unified description of particle & interaction
⟶ Standard Model
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⟶ Standard Model
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Coherent QM description of EM, Weak & Strong interactions…

…based on fundamental constituents (quarks & leptons) interacting via force carriers-photons, bosons, gluons

Brief History of Electrodynamics

Modern Period (1900 – Present)

1980s **Grand Unification!!!!!**

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String Theory

Brief History of Electrodynamics

Modern Period (1900 – Present)

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Incorporate all the four forces into a single dynamical force

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Modern Period (1900 – Present)

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THEORY OF EVERYTHING!!!

All these stories are interesting to sit back and listen to

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… we have to patiently & rigorously work out the basic nitty-gritty step-by-step!

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Physics is a set of conceptual ideas which are expressed mathematically

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Hence, we shall start with Electrostatics…

But, before that…

Physics is a set of conceptual ideas which are expressed mathematically

To have deep understanding of physics, we need to have deep understanding of basic/advance mathematics

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